# Zephyr Sensor Fusion with MicroBlaze V on Kria KR260 and OpenAMP

***Detail Design Guide2- Zephyr Device Tree Creation and Building Zephyr App***

## A. Jumpstart on Zephyr Device Tree

One can follow this readme document from the repository to modify the pl.dtsi and system_top.dts and generating mbv32.dtsusing **`LOPPER_DTC_FLAGS`** command:

https://github.com/LogicTronixInc/MicroBlaze-V-Zephyr-Kria-OpenAMP/blob/main/Documentations/README.md#2---zephyr-device-tree

**Additionally, those steps are also mentioned below,**

### Creating Zephyr Device Tree

First activate the zephyr environment:

```
logictronix06@logictronix06:~/zephyrproject$ source ~/zephyrproject/.venv/bin/activate
(.venv) logictronix06@logictronix06:~/zephyrproject$ cd zephyr
```

One can run the looper command to generate the Zephyr device tree:

```
LOPPER_DTC_FLAGS="-b 0 -@" west lopper-command -p microblaze_riscv_0 -
s <path_of_device_tree>/zephyr/zephyr/system-top.dts -w .
```

Below is the screenshot covering the looper command output:

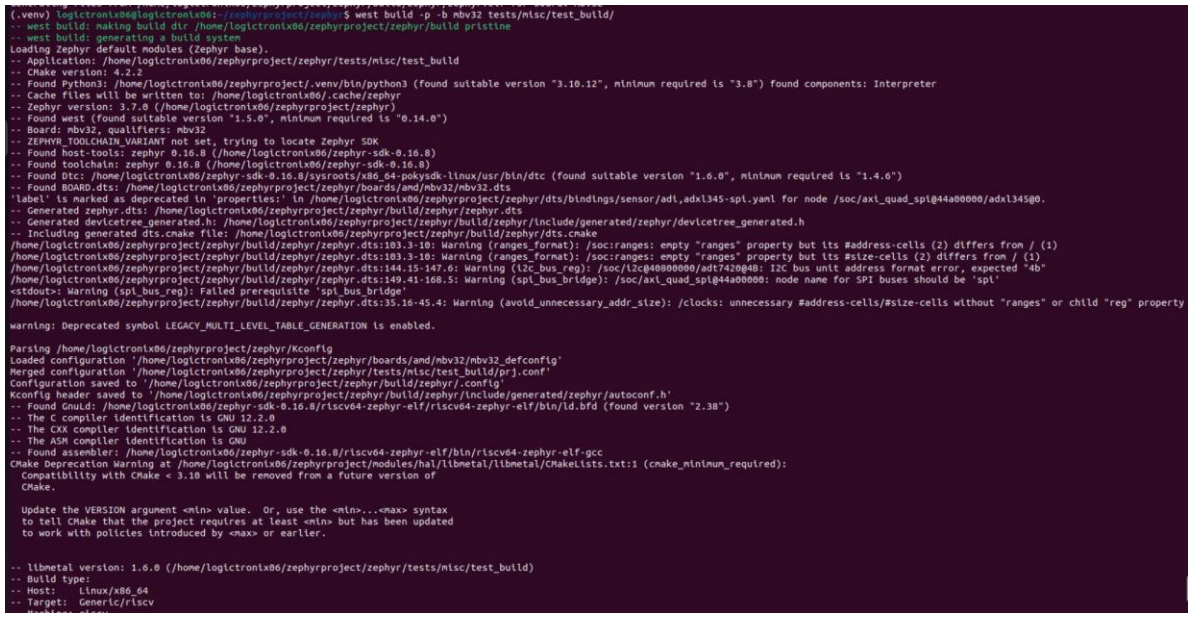
```
(.venv) logictronix06@logictronix06:~/zephyrproject/zephyr$ LOPPER_DTC_FLAGS="-b 0 -@" west lopper-command -p microblaze_riscv_0 -s /home/logictronix06/Videos/25_26_prj/Zephyr-Op
enAMP-MBlze-KR260/device_tree/zephyr/system-top.dts -w ~/zephyrproject/zephyr
{'__dbg__': 0,
 '__modified__': False,
 '__pstate__': 'resolved',
 'abs_path': '/cpus_microblaze_riscv@0/cpu@0/xlnx,data-size',
 'binary': False,
 'name': 'xlnx,data-size',
 'node': <lopper.tree.LopperNode object at 0x757a3754a750>,
 'number': -1,
 'pclass': <class 'list'>,
 'phandle_resolution': True,
 'ptype': <LopperFmt.UINT32: 8>,
 'string_val': 'xlnx,data-size = <0x20>;',
 'value': [32]}
(.venv) logictronix06@logictronix06:~/zephyrproject/zephyr$
```

After obtaining the mbv32.dts, now one can add the GPIO node and SPI node on the device tree as mentioned at the readme linked above.

## B. Building Zephyr Application

One can run following command to build the Zephyr application:

```
west build -p -b mbv32 tests/misc/test_build/
```



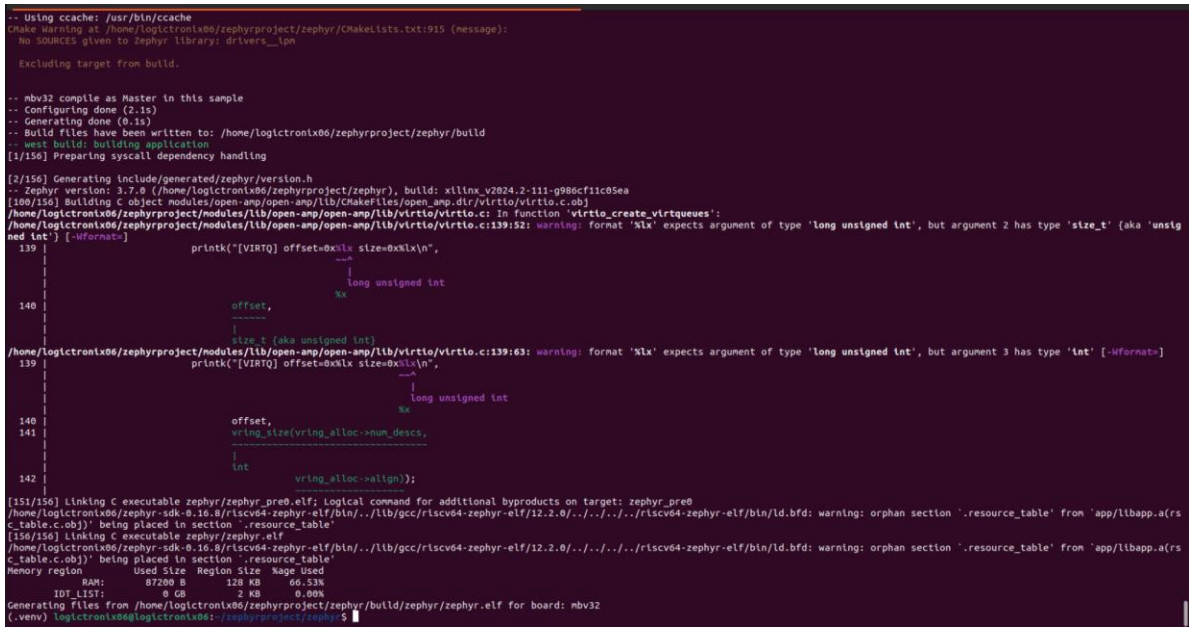
```
(.venv) logictronix06@logictronix06:~/zephyrproject/zephyr$ west build -p -b mbv32 tests/misc/test_build/
-- west build: making build dir /home/logictronix06/zephyrproject/zephyr/build pristine
-- west build: generating a build system
Loading Zephyr default modules (Zephyr base).
-- Application: /home/logictronix06/zephyrproject/zephyr/tests/misc/test_build
-- CMake version: 4.2.2
-- Found Python3: /home/logictronix06/zephyrproject/.venv/bin/python3 (found suitable version "3.10.12", minimum required is "3.8") found components: Interpreter
-- Cache files will be written to: /home/logictronix06/.cache/zephyr
-- Zephyr version: 3.7.0 (/home/logictronix06/zephyrproject/zephyr)
-- Found west (found suitable version "1.5.0", minimum required is "0.14.0")
-- Board: mbv32, qualifiers: mbv32
-- ZEPHYR_TOOLCHAIN_VARIANT not set, trying to locate Zephyr SDK
-- Found host-tools: zephyr 0.16.8 (/home/logictronix06/zephyr-sdk-0.16.8)
-- Found toolchain: zephyr 0.16.8 (/home/logictronix06/zephyr-sdk-0.16.8)
-- Found Dtc: /home/logictronix06/zephyr-sdk-0.16.8/sysroots/x86_64-pokysdk-linux/usr/bin/dtc (found suitable version "1.6.0", minimum required is "1.4.6")
-- Found BOARD.dts: /home/logictronix06/zephyrproject/zephyr/boards/amd/mbv32/mbv32.dts
'label' is marked as deprecated in 'properties:' in /home/logictronix06/zephyrproject/zephyr/dts/bindings/sensor/adi,adxl345-spi.yaml for node /soc/axi_quad_spi@44a00000/adxl345@0.
-- Generated zephyr.dts: /home/logictronix06/zephyrproject/zephyr/build/zephyr/zephyr.dts
-- Generated devicetree_generated.h: /home/logictronix06/zephyrproject/zephyr/build/zephyr/include/generated/zephyr/devicetree_generated.h
-- Including generated dts.cmake file: /home/logictronix06/zephyrproject/zephyr/build/zephyr/dts.cmake
/home/logictronix06/zephyrproject/zephyr/build/zephyr/zephyr.dts:103.3-10: Warning (ranges_format): /soc:ranges: empty "ranges" property but its #address-cells (2) differs from / (1)
/home/logictronix06/zephyrproject/zephyr/build/zephyr/zephyr.dts:103.3-10: Warning (ranges_format): /soc:ranges: empty "ranges" property but its #size-cells (2) differs from / (1)
/home/logictronix06/zephyrproject/zephyr/build/zephyr/zephyr.dts:144.15-147.6: Warning (i2c_bus_reg): /soc/i2c@40800000/adt7420@48: I2C bus unit address format error, expected "4b"
/home/logictronix06/zephyrproject/zephyr/build/zephyr/zephyr.dts:149.41-168.5: Warning (spi_bus_bridge): /soc/axi_quad_spi@44a00000: node name for SPI buses should be 'spi'
<stdout>: Warning (spi_bus_reg): Failed prerequisite 'spi_bus_bridge'
/home/logictronix06/zephyrproject/zephyr/build/zephyr/zephyr.dts:35.16-45.4: Warning (avoid_unnecessary_addr_size): /clocks: unnecessary #address-cells/#size-cells without "ranges" or child "reg" property

warning: Deprecated symbol LEGACY_MULTI_LEVEL_TABLE_GENERATION is enabled.

Parsing /home/logictronix06/zephyrproject/zephyr/Kconfig
Loaded configuration '/home/logictronix06/zephyrproject/zephyr/boards/amd/mbv32/mbv32_defconfig'
Merged configuration '/home/logictronix06/zephyrproject/zephyr/tests/misc/test_build/prj.conf'
Configuration saved to '/home/logictronix06/zephyrproject/zephyr/build/zephyr/.config'
Kconfig header saved to '/home/logictronix06/zephyrproject/zephyr/build/zephyr/include/generated/zephyr/autoconf.h'
-- Found GnuLd: /home/logictronix06/zephyr-sdk-0.16.8/riscv64-zephyr-elf/riscv64-zephyr-elf/bin/ld.bfd (found version "2.38")
-- The C compiler identification is GNU 12.2.0
-- The CXX compiler identification is GNU 12.2.0
-- The ASM compiler identification is GNU
-- Found assembler: /home/logictronix06/zephyr-sdk-0.16.8/riscv64-zephyr-elf/bin/riscv64-zephyr-elf-gcc
CMake Deprecation Warning at /home/logictronix06/zephyrproject/modules/hal/libmetal/libmetal/CMakeLists.txt:1 (cmake_minimum_required):
  Compatibility with CMake < 3.10 will be removed from a future version of
  CMake.

  Update the VERSION argument <min> value.  Or, use the <min>...<max> syntax
  to tell CMake that the project requires at least <min> but has been updated
  to work with policies introduced by <max> or earlier.


-- libmetal version: 1.6.0 (/home/logictronix06/zephyrproject/zephyr/tests/misc/test_build)
-- Build type:
-- Host:    Linux/x86_64
-- Target:  Generic/riscv
```



```
-- Using ccache: /usr/bin/ccache
CMake Warning at /home/logictronix06/zephyrproject/zephyr/CMakeLists.txt:915 (message):
  No SOURCES given to Zephyr library: drivers__lpm

  Excluding target from build.


-- mbv32 compile as Master in this sample
-- Configuring done (2.1s)
-- Generating done (0.1s)
-- Build files have been written to: /home/logictronix06/zephyrproject/zephyr/build
-- west build: building application
[1/156] Preparing syscall dependency handling

[2/156] Generating include/generated/zephyr/version.h
-- Zephyr version: 3.7.0 (/home/logictronix06/zephyrproject/zephyr), build: xilinx_v2024.2-111-g986cf11c05ea
[100/156] Building C object modules/open-amp/open-amp/lib/CMakeFiles/open_amp.dir/virtio/virtio.c.obj
/home/logictronix06/zephyrproject/modules/lib/open-amp/open-amp/lib/virtio/virtio.c: In function 'virtio_create_virtqueues':
/home/logictronix06/zephyrproject/modules/lib/open-amp/open-amp/lib/virtio/virtio.c:139:52: warning: format '%lx' expects argument of type 'long unsigned int', but argument 2 has type 'size_t' (aka 'unsig
ned int') [-Wformat=]
  139 |                         printk("[VIRTQ] offset=0x%lx size=0x%lx\n",
      |                                                   ~~^
      |                                                     |
      |                                                     long unsigned int
      |                                                   %x
  140 |                                 offset,
      |                                 ~~~~~~
      |                                 |
      |                                 size_t (aka unsigned int)
/home/logictronix06/zephyrproject/modules/lib/open-amp/open-amp/lib/virtio/virtio.c:139:63: warning: format '%lx' expects argument of type 'long unsigned int', but argument 3 has type 'int' [-Wformat=]
  139 |                         printk("[VIRTQ] offset=0x%lx size=0x%lx\n",
      |                                                         ~~^
      |                                                           |
      |                                                           long unsigned int
      |                                                         %x
  140 |                                 offset,
  141 |                                 vring_size(vring_alloc->num_descs,
      |                                 ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
      |                                 |
      |                                 int
  142 |                                         vring_alloc->align));
      |                                         ~~~~~~~~~~~~~~~~~~~~
[151/156] Linking C executable zephyr/zephyr_pre0.elf; Logical command for additional byproducts on target: zephyr_pre0
/home/logictronix06/zephyr-sdk-0.16.8/riscv64-zephyr-elf/bin/../lib/gcc/riscv64-zephyr-elf/12.2.0/../../../../riscv64-zephyr-elf/bin/ld.bfd: warning: orphan section `.resource_table' from `app/libapp.a(rs
c_table.c.obj)' being placed in section `.resource_table'
[156/156] Linking C executable zephyr/zephyr.elf
/home/logictronix06/zephyr-sdk-0.16.8/riscv64-zephyr-elf/bin/../lib/gcc/riscv64-zephyr-elf/12.2.0/../../../../riscv64-zephyr-elf/bin/ld.bfd: warning: orphan section `.resource_table' from `app/libapp.a(rs
c_table.c.obj)' being placed in section `.resource_table'
Memory region         Used Size  Region Size  %age Used
             RAM:       87200 B       128 KB     66.53%
        IDT_LIST:          0 GB         2 KB      0.00%
Generating files from /home/logictronix06/zephyrproject/zephyr/build/zephyr/zephyr.elf for board: mbv32
(.venv) logictronix06@logictronix06:~/zephyrproject/zephyr$
```

At the end of Zephyr west build, one will get the ***zephyr.elf*** file, this is the application file which we will be using in upcoming step.

In general, we will merge this ***zephyr.elf*** with the BIT file and create a single BIT only, so that we can load the BIT by renaming it to BIN at Kria Firmware load with ***xmutil***.

## Revision History

| S.N | Revision Number | Revision Details | Author |
|---|---|---|---|
| 1. | v1.0 | Initial Revision | Krishna Gaihre and Sanam Shakya |